Radio Frequency Signal Source Separation

Jared Miller
University of Utah

Tyler Trotter
University of Utah

Hyrum Bailey
University of Utah

***Abstract*—Signal Source Separation (SSS) in the radio frequency (RF) domain is a challenging problem due to overlapping signals, modulation types, and random noise. In this work, we study RF source separation under a simplified linear mixture model of the form $m = s_{\rm soi} + \alpha s_{\rm int} + \varepsilon$, inspired by the MIT RF Challenge [1], where $s_{\rm soi}$ is a signal of interest (SOI), $s_{\rm int}$ is an interfering wave signal, and ε represents additive noise in the system. Using synthetically generated quadrature phase shift keying (QPSK) signals, we construct both single-channel and multi-channel separation scenarios at different noise levels, phase, and interference strengths. Under these assumptions, we explored several deep learning approaches, compared against one another and to a FastICA baseline, including linear separation, convolutional and recurrent neural networks, and a state-of-the-art audio source separation model (HTDemucs) [2]. Performance is assessed using both waveform-level metrics, such as mean-squared error (MSE) and signal-to-distortion ratio (SDR), as well as symbol recovery. Our results point to a distinction between waveform fidelity and symbol accuracy: in the single-channel case, all tested models and FastICA retained perfect symbol recovery across the tested sweep range, so waveform metrics provided the more useful comparison. In the two-channel case, giving FastICA multiple observations improved the baseline substantially, but learned models still achieved lower waveform error. These findings motivate future evaluation in harder multi-receiver settings where symbol recovery becomes a more discriminative measure.**

I. Introduction

Signal source separation (SSS) is a fundamental problem in signal processing with broad applications spanning audio engineering, wireless communications, and defense systems. In its classical form, SSS seeks to recover individual latent signals from an observed mixture. Well-known examples include music source separation, where individual instruments are extracted from a composite recording, and interference mitigation in communication systems, where unwanted signals must be removed to recover a desired transmission.

In the radio frequency (RF) domain, the problem becomes particularly challenging due to the diversity of modulation schemes, channel effects, and the presence of structured as well as stochastic interference. Practical applications include anti-jamming in contested environments, spectrum monitoring, and improving receiver robustness across heterogeneous signal environments. For instance, in maritime or defense settings, the ability to isolate a signal of interest (SOI) from interfering transmissions can directly impact communication reliability and operational effectiveness.

In this work, we consider a simplified but representative linear mixture model inspired by the MIT Radio Frequency Challenge [1]:

$$s_{\rm mix} = s_{\rm soi} + \alpha s_{\rm int} + \varepsilon \tag{1}$$

where $s_{\rm mix}$ denotes the observed mixture, $s_{\rm soi}$ is the signal of interest (SOI), $s_{\rm int}$ represents an interfering signal, $\alpha \in \mathbb{R}^+$ is a scaling factor controlling interference strength, and ε models additive zero-mean Gaussian noise. This formulation captures the core difficulty of RF source separation: disentangling overlapping signals in the presence of both structured interference and random noise.

A key question in this setting is the level at which separation should be evaluated. In many communication systems, the ultimate objective is not the waveform itself but the accurate recovery of transmitted symbols. This motivates a comparison between waveform-level separation and *symbol recovery*, where performance is measured by the correctness of decoded information, rather than signal clarity alone.

II. Background

Both $s_{\rm soi}$ and $s_{\rm int}$ are generated as QPSK-modulated complex baseband signals,

$$s(t) = I(t) + jQ(t), \tag{2}$$

where $I(t)$ is the in-phase component and $Q(t)$ is the quadrature component. Each transmitted symbol is selected from a four-point QPSK constellation. After pulse shaping with a root-raised cosine (RRC) filter, the symbol sequence becomes a continuous-time waveform that can be shifted to a carrier frequency and transmitted. At the receiver, demodulation, matched filtering, and sampling recover estimates of the original encoded symbols [3].

The final symbol decision step is important in this work because waveform reconstruction and symbol recovery do not always agree. A separator can produce a visibly imperfect waveform while still recovering the correct symbol stream, or it can reconstruct a smoother waveform while making poorer hard symbol decisions.

III. Method

The system architecture separates training-time and test-time workflows. The data generator creates synthetic mixtures, the separation models estimate the individual sources, and a symbol recovery stage converts separated waveforms back to QPSK decisions. The non-trivial parts of the system are the data generator, the separation models, and the evaluation framework.

A. Data Generation

A custom RF data generator was developed to support this work because no directly usable open-source dataset matched the simplified mixture model and experimental controls needed for the project. The generator constructs QPSK waveforms, applies pulse shaping, mixes the SOI and interference using the chosen value of α, optionally adds Gaussian noise using a target SNR in dB, and can generate either a single received channel or a multi-receiver observation. The same generator also allows the number of symbols, pulse-shape parameters, normalization behavior, and number of receive channels to be controlled from the experiment configuration.

B. Source Separation

The repo supports several learned source separators together with a FastICA baseline. In this work we focus on Linear, Hybrid, LSTM, IQ_CNN, HTDemucs, and FastICA. The learned models all operate on I/Q channelized inputs, while FastICA provides a classical blind source separation reference.

IV. Models Used, Sources, and Modifications

To solve the RF source separation problem we used five learned architectures and one classical baseline. The learned architectures were Linear, Hybrid, LSTM, IQ_CNN, and HTDemucs [4]. All architectures were adapted to fit the project data generation, training, and validation pipeline.

A. FastICA

In signal processing, independent component analysis (ICA) is a computational method for separating a multivariate signal into additive subcomponents. By assumption, at most one subcomponent is Gaussian and the underlying sources are mutually independent [5]. FastICA iteratively searches for directions that maximize non-Gaussianity, which is used as a proxy for source independence. In this project, FastICA is used as a non-neural baseline for blind source separation.

B. Hybrid Separator

The hybrid separator combines a linear front-end with a deeper temporal network. It begins with an input projection layer implemented as a 1×1 convolution that mixes the I and Q channels into a learned feature vector. It then applies a sequence of temporal convolutional blocks with kernel size 9, which allows the network to learn local and medium-range structure over time. A final 1×1 projection maps the learned representation back into the desired separated outputs.

C. LSTM Separator

The LSTM separator was adapted from the sequence model provided by Professor Tom Henderson. For each example, the I and Q values across the receive channels are organized into a time-major sequence and passed through an input projection followed by a bidirectional LSTM. This lets the model learn dependencies from both the past and the future of the current sample. A small amount of dropout was incorporated between layers for better generalization.

D. IQ_CNN Separator

The IQ_CNN separator is a U-Net style temporal convolutional model designed for RF source separation. Unlike the linear separator, which only applies a pointwise channel projection, the IQ_CNN learns both local temporal structure and multi-scale features through an encoder–decoder architecture. The model first reshapes the input so that the I and Q components for each receive channel are grouped together, then applies a 1×1 projection to mix the antenna/IQ channels into a learned feature space. The encoder consists of three downsampling convolutional blocks, which reduce the time resolution while increasing the feature depth. At the bottleneck, the network applies dilated convolutions with dilation rates of 1, 2, and 4, increasing the receptive field without additional downsampling. The decoder then upsamples the signal back to the original resolution using transposed convolutions and skip connections, allowing the model to preserve fine waveform detail while still using the larger-scale context learned in the bottleneck. Group normalization [6] and GELU activations [7] are used throughout the network, and dropout can be included for regularization. In practice, this model acts as a stronger nonlinear separator than the linear baseline and is intended to better capture the temporal and structural patterns of RF waveforms.

E. HTDemucs Separator

The Hybrid Transformer Deep Extractor for Music Sources (HTDemucs) is a state-of-the-art open-source audio source separation model developed by Meta Research (Meta AI). The neural architecture baseline of Demucs is that of a Wave-U-Net, including a convolutional encoder stream, a central bidirectional long short-term memory bottleneck, and a convolutional decoder with transposed convolutions [2]. Given Demucs' success on waveform-based audio source separation, this work also adapted the architecture to RF separation.

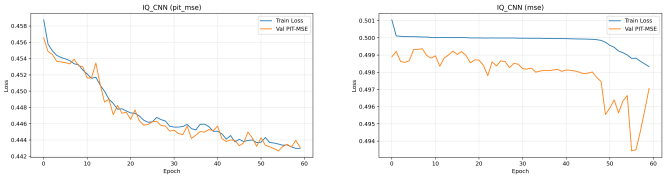
V. Training

Training proceeded in two stages. First, the project compared multiple model families under a shared training pipeline. The early comparison stage established IQ_CNN and LSTM as the strongest candidates for further study, with IQ_CNN slightly ahead in the earlier comparison runs. Second, we used cross-validation and targeted hyperparameter search to tune the IQ_CNN family.

The first cross-validation stage used a broad 12-hour search over five folds. This search varied base channel width, dropout, learning rate, weight decay, input noise, receiver dropout, gradient clipping, scheduler type, batch size, and normalization behavior. The best broad-search configuration used base_channels $= 48$, dropout $= 0.08$, learning rate $= 1.6887 \times 10^{-3}$, weight decay $= 1.315 \times 10^{-4}$, input-noise standard deviation $= 0.005$, receiver dropout probability $= 0.1$, gradient clipping $= 2.0$, and a plateau scheduler. That run produced a mean validation PIT-MSE of approximately 0.4510 and identified a stable parameter region for the IQ_CNN architecture.

TABLE I: Summary of the two cross-validation stages used to tune the IQ_CNN family.

Stage	Best Val Metric	Main Outcome
Broad CV	0.4510 PIT-MSE	Identified best parameter region
Focused CV	0.4460 PIT-MSE	Produced final tuned IQ_CNN setting



(a) PIT-MSE training study (b) Plain MSE training study

Fig. 1: Two-channel IQ_CNN loss study comparing PIT-MSE and plain MSE on the same fixed dataset. PIT-MSE learned more cleanly in waveform space, while plain MSE produced only a negligible edge in symbol accuracy.

The second stage used a focused deep search around that promising region. Instead of exploring a large search space again, the search fixed the general architecture around base_channels $= 48$ and trained a smaller number of configurations for up to 200 epochs. The best focused configuration retained dropout $= 0.08$, learning rate $= 1.6887 \times 10^{-3}$, weight decay $= 1.315 \times 10^{-4}$, input noise $= 0.005$, receiver dropout $= 0.1$, and the plateau scheduler. This improved the best validation PIT-MSE to roughly 0.4460 and produced the strongest tuned IQ_CNN configuration.

For the two-channel IQ_CNN study, we also compared two training losses on the same fixed two-receiver dataset with 20,000 training examples, 2,000 validation examples, and 2,000 test examples. The PIT-MSE objective learned more effectively in waveform space, reaching a best validation PIT-MSE of 0.4427 and a test SDR of 0.52 dB. A plain MSE loss gave slightly higher average symbol accuracy on that specific study (0.6292 versus 0.6286), but at noticeably worse waveform quality. Because the symbol gap was extremely small while the waveform gap was much larger, PIT-MSE remained the more principled separation loss, while plain MSE was retained as an alternate option for further symbol-focused experiments.

VI. Experiments

This report details experiments for the single-channel, two-channel, and four-channel separation problems. The number of channels corresponds to the number of receive antennas observing the transmitted mixture. As the number of channels increases, the problem changes from a simple instantaneous mixture to a more realistic multi-observation separation task.

A. A. Single-Channel Separation Problem

The first finalized reporting run focused on the single-channel case, where one complex received waveform was represented as a two-channel I/Q input. In this setting, the project evaluated

TABLE II: Single-channel reference comparison at $\alpha = 1.0$ and SNR $= 15$ dB.

Model	Wave MSE	SDR (dB)	SOI Acc
LSTM	0.0210	14.17	1.000
IQ_CNN	0.0240	13.51	1.000
Hybrid	0.0319	12.20	1.000
Tiny	0.0755	8.26	1.000
Linear	0.2589	2.88	1.000
FastICA	0.6670	-1.16	1.000

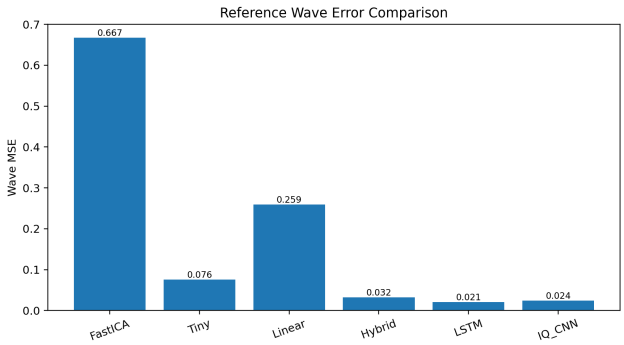


Fig. 2: Single-channel reference waveform comparison. Since symbol accuracy was saturated, waveform error was the main discriminative metric.

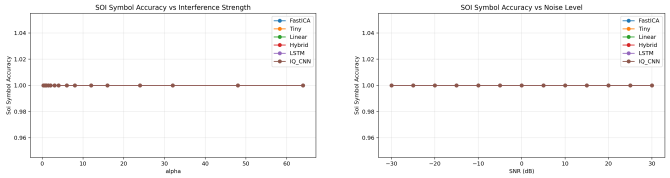
FastICA, Tiny, Linear, Hybrid, LSTM, and IQ_CNN using a fixed reference dataset together with alpha and noise sweeps.

At the reference point ($\alpha = 1.0$, SNR $= 15$ dB), all tested methods achieved 100% symbol accuracy for both the SOI and the interference. The main distinction between methods therefore came from waveform quality. LSTM and IQ_CNN produced the lowest wave error, followed by Hybrid. Linear and FastICA were much weaker in waveform reconstruction, even though they still recovered the symbol stream correctly. This was one of the most important findings of the single-channel study: in easier conditions, waveform metrics are often more informative than symbol metrics because symbol accuracy can remain saturated.

The alpha and noise sweeps reinforced this conclusion. Across the tested single-channel sweep range, none of the models dropped below the 95% symbol-accuracy threshold, and no alpha or SNR failure point was observed. However, the waveform plots clearly separated the stronger learned models from the weaker baselines.

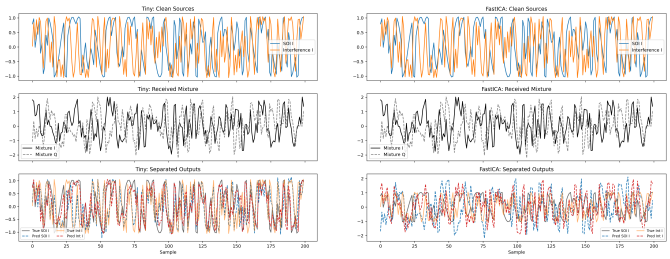
B. B. 2-Channel Separation Problem

The second major experiment moved to two receive channels. This step was especially important for evaluating FastICA fairly, because ICA is fundamentally a multivariate blind source separation method and benefits from multiple observations. A preliminary validation step compared FastICA in the one-channel and two-channel settings. Moving from one complex receive channel to two receive channels improved FastICA dramatically: wave MSE decreased from 0.3824 to 0.0660, SDR increased from 1.17 dB to 10.76 dB, and symbol accuracy



(a) SOI accuracy vs. α (b) SOI accuracy vs. SNR

Fig. 3: Single-channel robustness sweeps. Symbol recovery remained saturated across the tested range, confirming that this setting was still relatively easy from the standpoint of hard symbol decisions.



(a) Best learned model example (b) FastICA example

Fig. 4: Representative single-channel examples showing that waveform quality can differ noticeably even when all methods still recover the correct symbol stream.

improved from about 40% to approximately 97.3% for both the SOI and the interference.

After establishing that FastICA needed two receive channels to act as a fairer baseline, we compared FastICA against the learned models on the same fixed two-channel dataset generated through the main project generator. In this reference comparison, FastICA remained the weakest method by a large margin in waveform quality, while the learned models clustered more closely together. LSTM produced the highest symbol accuracy among the learned models, but the gap between the learned models was small. In contrast, FastICA's wave MSE remained much higher than all learned models. This result supports the view that a multichannel FastICA baseline is much fairer than the single-channel baseline, but still not competitive with the learned separators in waveform recovery.

C. C. 4-Channel Separation Problem

For the four-channel separation problem, a simple mean squared error (MSE) loss was optimized using Adam [8]. All models were trained on 10,000 training examples and validated on 1,000 examples for 300 epochs. In this setting, none of the created models generalized well to the validation set. The longer the models trained, the more they overfit to the training data, which is visible in Fig. 8. This result motivated the single-channel and two-channel experiments described above, which were designed to understand whether the pipeline could learn under simpler but still structured RF conditions.

TABLE III: Two-channel reference comparison using the main repo generator.

Model	Wave MSE	SDR (dB)	SOI Acc
LSTM	0.4570	0.42	0.631
Hybrid	0.4548	0.44	0.629
IQ_CNN	0.4544	0.44	0.626
Linear	0.4600	0.39	0.625
Tiny	0.4567	0.43	0.622
FastICA	0.8763	-2.32	0.621

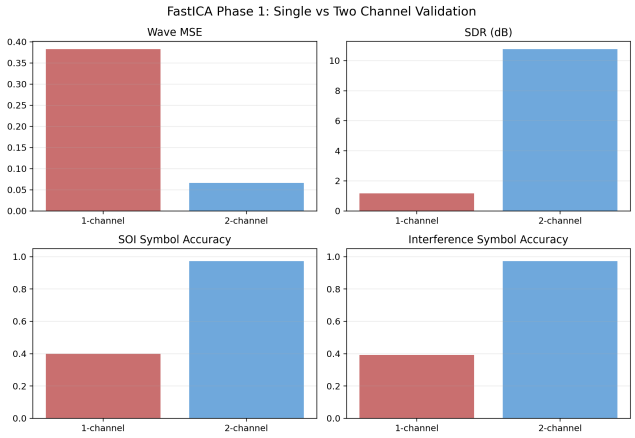


Fig. 5: FastICA Phase 1 validation: moving from one receive channel to two receive channels greatly improves the baseline. This confirms that ICA should not be treated as a meaningful one-channel baseline for the current problem.

VII. Conclusion

This work produced three main lessons. First, in the easier single-channel RF setting, all models and even the FastICA baseline retained perfect symbol recovery across the tested range, so waveform metrics were the more informative comparison tool. Second, FastICA is not a meaningful one-channel baseline for this problem, but it becomes much more reasonable when given two receive channels. Third, even after giving FastICA a fairer multichannel setting, the learned models still achieved better waveform recovery. These results justify future work on more difficult two-channel and four-channel settings where symbol recovery can become a stronger discriminator between models.

References

[1] Massachusetts Institute of Technology, "Rf challenge: Ai accelerator," https://rfchallenge.mit.edu/, 2026, accessed 2026.

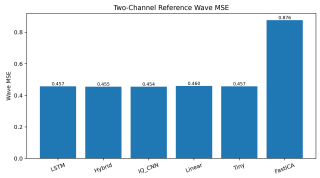
[2] A. Défossez, N. Usunier, L. Bottou, and F. Bach, "Demucs: Deep extractor for music sources with extra unlabeled data remixed," *arXiv preprint arXiv:1909.01174*, 2019.

[3] B. Sklar, *Digital Communications: Fundamentals and Applications*, 2nd ed. Prentice Hall PTR, 2000.

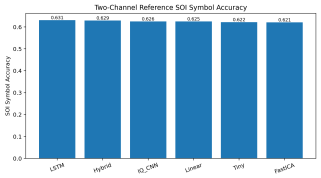
[4] Facebook AI Research, "Demucs: Music source separation," https://github.com/facebookresearch/demucs, 2026, gitHub repository.

[5] Wikipedia contributors, "Independent component analysis," https://en.wikipedia.org/wiki/Independent_component_analysis, 2026, accessed 2026-04-07.

[6] Y. Wu and K. He, "Group normalization," *arXiv preprint arXiv:1803.08494*, 2018.

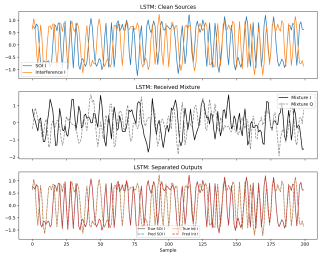


(a) Wave MSE comparison

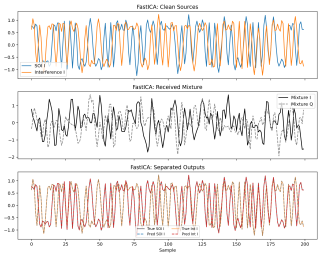


(b) SOI symbol accuracy comparison

Fig. 6: Two-channel reference comparison. The learned models are similar in symbol accuracy, but FastICA remains much worse in waveform quality.



(a) Best learned model example



(b) FastICA example

Fig. 7: Representative two-channel outputs. Even with two observations, the learned models reconstruct cleaner waveforms than the FastICA baseline.

[7] D. Hendrycks and K. Gimpel, "Gaussian error linear units (gelus)," *arXiv preprint arXiv:1606.08415*, 2016.

[8] D. P. Kingma and J. Ba, "Adam: A method for stochastic optimization," *arXiv preprint arXiv:1412.6980*, 2014.

[9] N. Takahashi and Y. Mitsufuji, "D3net: Densely connected multi-dilated densenet for music source separation," *arXiv preprint arXiv:2010.01733*, 2020.

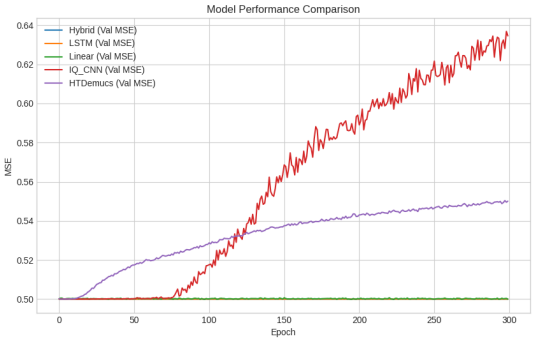


Fig. 8: Validation MSE for the four-channel experiments after 300 epochs. The figure shows strong overfitting and poor validation generalization in the most difficult setting.

Appendix A
Music Source Separation

At the start of the semester our sponsor tasked us with Music Source Separation (MSS). Music source separation has evolved rapidly with the introduction of deep learning. Early methods relied on non-negative matrix factorization and handcrafted features, which were limited in their ability to disentangle overlapping sources. MSS takes as input a song and attempts to isolate specific parts of that song. The most common dataset used in the field is MUSDB18. This dataset contains 150 full-length music tracks as well as their isolated parts. MUSDB18 contains isolated drums, bass, voice, and an other category for everything else. The sponsor recommended that we use the D3Net architecture [9]. After reviewing the literature and benchmark results, the team transitioned from D3Net toward Demucs because Demucs represented a stronger state-of-the-art waveform separator for the sponsor's original problem.

Appendix B
MIT Interference Challenge

This project was ultimately shaped by the MIT RF Challenge framing, which provided the motivating mixture model and the broader source-separation objective [1]. The experiments in this report should be interpreted as controlled steps toward that broader interference-separation problem rather than as the final challenge solution itself.